MIL-E-1/1610C(EC)
4 June 1976
SUPERSEDING
MIL-E-1/1610B(NAVY)
30 August 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 8639

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Beam power tetrode, high voltage series regulator

DIMENSIONS AND PIN CONNECTIONS: See figure 1

CATHODE: Oxide coated

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Ib	Ic1	Pg1	Pg2	Pp	TE	tk	Alt
Unit:	V	Vdc	Vdc	Vdc	v	mAdc	mAdc	W	W	W	°C	sec	ft
Maximum:	6.9/13.8	4,000	-175	450	100	2x110	10	2	7	40	250	---	Note 1
Minimum:	5.7/11.4	---	---	---	---	---	---	---	---	---	---	60	---
TEST CONDITIONS:	6.3/12.6	600	Adj	250	---	57	---	---	---	---	---	120	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) - 72 hours

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-	Ef = 6.3 V; t = 300	1.0	II	If	1.6	2.0	A
1256	Electrode current (screen)	2		1.0	II	Ic2	---	9	mAdc
1261	Electrode voltage (1) (control)	2		1.0	II	Ec1	-18	-29	Vdc
1261	Electrode voltage (2) (control)	2, 3	Ebb = 5,000 Vdc; Rp = 3 Meg; Ec2 = 450 Vdc; Ec1/Ib = 1 mAdc	1.0	II	Ec1	-56	-87	Vdc
1266	Total grid current	2, 4	t = 180	1.0	II	Ic1	---	-8.5	µAdc
1231	Peak emission	2	eb = ec1 = ec2 = 225v	1.0	II	is	3	---	a
4201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	2	Eb = 250 V; Ec1/Ib = 15 mAdc; Rp = 2,000 ohms	---	---	Ep	---	1,200	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	18	µAdc
1266	Primary grid emission (screen)	2	Pg2 = 7 W; t = 15 sec	---	---	Ig2	---	-450	µA
1101	Secureness of base	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	5	Group C; Ebb = 5,000 Vdc; Ecc2 = 130 Vdc; Rg2 = 1,000 ohms; Rp = 100,000 ohms; vary Rk/Ib = 10 mAdc	---	---	---	---	---	---
---	Life-test end point: (500 hours)								
1266	Total grid current	-		---	---	Ic1	---	-20	µA
1266	Primary grid emission (screen)	-		---	---	Ig2	---	-1.0	mA

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Tie 1a to 2a.
3. Use a 100 K ohm resistor paralleled by neon bulb in series with anode. Arcing during "hold off" will light neon bulb. Hold Ec1 at -150 Vdc until 1 minute elapses without arcing. Adjust Ec1 less negative until Ib is at the test conditions specified. Neon bulb will be lit under these conditions.
4. This test to be performed at the conclusion of the holding period.
5. For lots of 25 tubes or less, 1 tube per lot shall be life tested and the tube shall not fail during the life-test period specified.

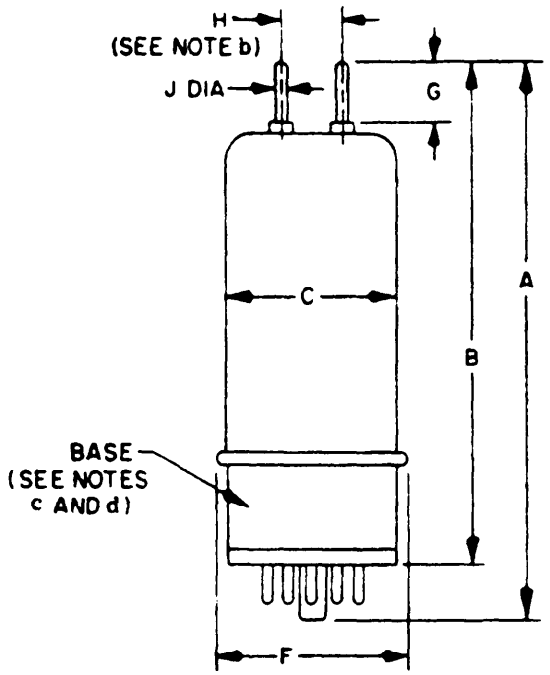
User activities:
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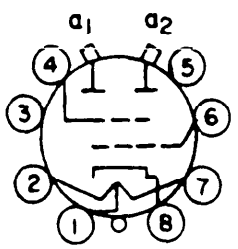
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.750(120.65)	5.090(129.29)
C		1.811 (46.00)
Quality conformance inspection, part 3 (see note e)		
B	4.240(107.70)	4.560(115.82)
F		1.938 (49.23)
G	.475 (12.07)	
H	.550(13.97) Nominal	
J	.077 (1.96)	.083 (2.11)

PIN CONNECTIONS

1 Heater, center tap
2 Heater
3 No connection
4 Grid No. 2
5 No connection
6 Grid No. 1
7 Heater
8 Cathode
a1 Anode (See note a)
a2 Anode (See note a)



NOTES:
a. Both anode pins shall be connected together externally. Connection between pins shall not be rigid.
b. The anode leads shall be capable of entering a flat plate gage of .375 ±.001 (9.53 ±.03 mm) thickness having two holes .1400 ±.0005 (3.56 ±.01 mm) in diameter spaced .550 ±.001 (13.97 ±.03 mm) apart.
c. Phenolic shell octal base B8-71, with external barriers.
d. The plane through the center of the key shall be 90 degrees nominal to plane through center of anode pins.
e. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8639.

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